Artificial Intelligence (AI) and Generative AI (GenAI)

Artificial Intelligence (AI)
AI is the field of computer science focused on building systems that can perform tasks requiring human intelligence, such as learning, reasoning, perception, language understanding, and decision-making.

Types of AI
- Narrow AI: Designed for a specific task (e.g., spam detection).
- General AI: Human-level intelligence across many tasks (not yet achieved).
- Super AI: Hypothetical intelligence beyond humans.

Common AI Techniques
Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Reinforcement Learning.

Generative AI (GenAI)
Generative AI is a branch of AI that creates new content such as text, images, audio, video, and code based on learned patterns from data.

How GenAI Works
1. Data collection and preprocessing.
2. Model training using neural networks (often Transformers).
3. Prompt input from the user.
4. Content generation and refinement.

Popular Models
GPT, Llama, Qwen, Gemini, Claude, Mistral, Stable Diffusion.

Applications
Chatbots, content writing, coding assistants, document summarization, translation, RAG systems, image generation, healthcare, education, and customer support.

Advantages
Automation, productivity, creativity, personalization, and faster decision support.

Limitations
Hallucinations, bias, privacy concerns, high computational cost, and dependence on training data.

AI vs GenAI
AI focuses on analyzing, predicting, and making decisions. GenAI focuses on creating new content from learned patterns.